

// PROJECT PROPOSAL // GROUP 45



WOMEN TECHSTERS BOOTCAMP BY TECH4DEV · COHORT 5.0

// GROUP 45 — CAPSTONE PROJECT — 2026

CYBER SHIELD

HUMAN FIREWALL CHALLENGE

AI Scam Simulator + Immunity Trainer

PROJECT TYPE

Capstone Project

TRACK

Cybersecurity + AI

SDG ALIGNMENT

SDG 4 // 9 // 16

PROJECT ONE-LINER

"Cyber Shield doesn't just detect scams — it trains users to resist them."

CYBER SHIELD

// 01 PROBLEM STATEMENT

The Problem We Are Solving

Online scams are becoming increasingly sophisticated, targeting individuals through phishing emails, fake websites, social engineering messages, and fraudulent links.

While many cybersecurity tools focus on detecting scams, the human factor remains the weakest link. Employees, students, children, and elderly individuals often fall victim due to lack of practical awareness and real-world exposure.

Organizations also face major financial and reputational losses due to human error in handling cyber threats.

There is a need for an interactive and practical training solution that helps users recognize and resist scams before real attackers exploit them.

// 03 PROPOSED SOLUTION

What Is Cyber Shield?

Cyber Shield is an AI-powered interactive scam simulation platform designed to train users to identify and resist cyber threats.

Rather than simply detecting scams, the platform builds user immunity through realistic, hands-on experience.

KEY FEATURES

- AI-generated phishing emails and scam messages
- Fake login pages to test credential safety awareness
- Interactive escape-room style challenges
- Real-time feedback on user decisions
- Personalized cybersecurity improvement tips

Users learn by doing, making cybersecurity training engaging, practical, and memorable.

CYBER SHIELD

Roles and Responsibilities

A. TECHNICAL PROJECT MANAGER

(Best fit for Technical PM students)

RESPONSIBILITIES

- Plan project scope and objectives
- Create timeline and sprint plans
- Assign responsibilities and manage workflow
- Coordinate communication between technical teams
- Track risks and ensure timely delivery
- Prepare project documentation and reports

DELIVERABLES

- // Project roadmap
- // Task schedule
- // Risk management plan
- // Progress tracking dashboard

B. CYBERSECURITY ANALYST

(Cybersecurity students)

RESPONSIBILITIES

- Research modern cyber threats and scam patterns
- Design realistic phishing and scam scenarios
- Develop cybersecurity awareness content
- Identify user vulnerabilities
- Ensure platform follows security best practices

DELIVERABLES

- // Scam scenario database
- // Threat analysis report
- // User safety guidelines

C. SECURITY OPERATIONS AND RISK LEAD

(Cybersecurity students)

RESPONSIBILITIES

- Design the Human Risk Score framework
- Analyze how users respond to threats
- Map behavioral weaknesses
- Develop mitigation strategies
- Ensure ethical handling of user data

DELIVERABLES

- // Risk scoring model
- // Behavior analysis framework
- // Security compliance checklist

// 04 TEAM TASK SHARING (CONTINUED)

D. AI SYSTEMS COORDINATOR

(Cybersecurity student with coding OR Technical PM with tech skills)

RESPONSIBILITIES

- Plan AI feature implementation
- Coordinate between AI development and platform team
- Oversee scam scenario automation
- Manage AI workflow and testing
- Ensure AI outputs align with project goals

DELIVERABLES

- // AI feature plan
- // AI workflow diagrams
- // System integration documentation

E. TECHNICAL IMPLEMENTATION LEAD

(Cybersecurity student with coding background)

RESPONSIBILITIES

- Develop interactive simulation platform
- Build phishing email and fake login interfaces
- Integrate AI and scoring systems
- Manage backend functionality
- Ensure system reliability

DELIVERABLES

- // Functional application
- // Simulation engine
- // System integration

F. DOCUMENTATION AND PRESENTATION LEAD

(Technical PM students)

RESPONSIBILITIES

- Prepare project proposal and documentation
- Develop demo presentation slides
- Translate technical features into simple explanations
- Coordinate demo storytelling
- Prepare user guides

DELIVERABLES

- // Project documentation
- // Presentation slides
- // Demo script

SHARED RESPONSIBILITIES

Brainstorming features // Testing simulations // Demo rehearsal // Feedback improvements

// 05 AI COMPONENT

Artificial Intelligence Functions

SCENARIO GENERATION

AI creates realistic phishing emails, scam messages, and fraudulent web pages.

BEHAVIORAL ANALYSIS

AI monitors how users interact with simulations — click patterns, hesitation time, information entered.

RISK SCORING SYSTEM

AI calculates a Human Risk Score indicating how vulnerable a user is to scams.

PERSONALIZED TRAINING

AI recommends tailored safety tips and learning modules based on user weaknesses.

// 06 PROJECT IMPACT

Who Does This Protect?

COMPANIES

Staff cybersecurity awareness and phishing resistance training.

SCHOOLS

Digital safety education integrated into curricula.

FAMILIES

Child online protection and parental scam guidance.

ELDERLY

Accessible scam awareness for the most vulnerable.